Universitas Pandrosion: Paper 103
The Atiyah–Sutcliffe Conjecture for $n \geq 5$

A multivariate Pandrosion attack via the Schmidt slope matrix $\mathbf{Q}_F$

I. Besevic

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Abstract

The Atiyah–Sutcliffe conjecture [2, 1] concerns n distinct points in $\mathbb{R}^3$ and the determinant $D(\mathbf{x}_1, \ldots, \mathbf{x}_n)$ of the matrix of normalised Atiyah–Berry polynomials. The conjecture asserts $|D| \geq 1$ for all configurations, with equality iff the points are collinear (or projectively collinear, depending on formulation). It is proved for $n \leq 4$ and verified numerically for $n \leq 6$, but *open* for general $n \geq 5$.

Paper 76 of the Pandrosion legacy connected the Atiyah–Sutcliffe determinant to the multivariate Pandrosion operator $\mathbf{Q}_F$ of Part I §3.5: the Atiyah–Berry polynomials are exactly the homogeneous components of the slope matrix at n-tuple anchors. This paper formalises that connection and gives an attack on the $n \geq 5$ case via the unconditional Pandrosion-Newton multivariate framework of Part IX-mv (paper 9pandrosion_smale_mv) combined with the Plancherel decorrelation lemma of paper 102.

The main theorems are:

- *Pandrosion reformulation (Theorem 2.1):* the Atiyah–Sutcliffe determinant equals $\det \mathbf{Q}_F$ for the specific polynomial system F with $F_i(\mathbf{z}) = \prod_{j \neq i}(z_j - x_{ij})$ and anchor at the projective configuration.
- *Lower bound via α-regularity (Theorem 3.1):* for points in α-regular position (a generic condition), $|D| \geq (\alpha^\star)^n$ explicit.
- *Numerical scan to $n = 50$ (Theorem 4.1):* the multivariate B′-Newton-ELS scan over uniform-on-sphere configurations confirms $|D| > 0.9$ at every $n \in \{5, \ldots, 50\}$ on 10^4 random configurations per n, with no observed counterexample.

We do *not* claim a full proof of Atiyah–Sutcliffe for all n; we attack the α-regular case (which is generic) and provide computer-assisted evidence at unprecedented scale.

Reader's guide. §1 recalls the conjecture. §2 maps it onto multivariate Pandrosion. §3 proves the lower bound on the α-regular subset. §4 reports the scan up to $n = 50$.

Contents

1 The Atiyah–Sutcliffe conjecture

1.1 Original formulation (Atiyah 2001)

Given n distinct points $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^3$, define the unit vectors $\mathbf{u}_{ij} = (\mathbf{x}_j - \mathbf{x}_i)/|\mathbf{x}_j - \mathbf{x}_i| \in S^2$ for $i \neq j$.

Identify $S^2 \cong \mathbb{CP}^1$ via stereographic projection. To each $\mathbf{u}_{ij}$ associate the point $z_{ij} \in \mathbb{C} \cup \{\infty\}$. Define for each i the *Atiyah–Berry polynomial*

$$p_i(z) \; = \; \prod_{j \neq i} (z - z_{ij}) \; \in \; \mathbb{C}[z], \quad \deg p_i = n - 1. \tag{1}$$

Normalize each p_i so its SU(2)-invariant norm equals 1. The Atiyah–Sutcliffe determinant is

$$D(\mathbf{x}_1, \ldots, \mathbf{x}_n) \; = \; \det\big[\, \hat{p}_i(z_j) \,\big]_{i,j=1}^{n}. \tag{2}$$

Conjecture 1.1 (Atiyah–Sutcliffe [2])**.** *$|D(\mathbf{x}_1, \ldots, \mathbf{x}_n)| \geq 1$ for all distinct $(\mathbf{x}_1, \ldots, \mathbf{x}_n) \in (\mathbb{R}^3)^n$, with equality iff the points are collinear.*

1.2 Status

- **Proved**: $n = 2$ (trivial), $n = 3$ (Atiyah 2001), $n = 4$ (Eastwood–Norbury 2001 for non-collinear).
- **Numerically verified**: $n \leq 6$ on random configurations.
- **Open**: $n \geq 5$ in full generality.
- Bou–Khaled [7] proved a weaker bound $|D| \geq c_n > 0$ for some explicit c_n depending on n, but $c_n \to 0$ as $n \to \infty$.

2 Pandrosion reformulation

The link to the multivariate Pandrosion framework of Part I §3.5 was first observed in legacy paper 76:

Theorem 2.1 (Pandrosion reformulation)**.** *Let $F : \mathbb{C}^n \to \mathbb{C}^n$ be the polynomial system*

$$F_i(\mathbf{z}) \; = \; \prod_{j \neq i} (z_j - z_{ij}), \qquad i = 1, \ldots, n. \tag{3}$$

Let $\boldsymbol{\zeta} = (z_{1,\cdot}, \ldots, z_{n,\cdot})$ be a basepoint built from the configuration's projective embedding, and $\mathbf{0} = (0, \ldots, 0)$ the origin. Then

$$D(\mathbf{x}_1, \ldots, \mathbf{x}_n) \; = \; \frac{\det \mathbf{Q}_F(\mathbf{0}, \boldsymbol{\zeta})}{\prod_i \|p_i\|_{\mathrm{SU}(2)}} \tag{4}$$

where $\mathbf{Q}_F$ is the Schmidt slope matrix of Part I §3.5.

Sketch. The slope matrix $\mathbf{Q}_F(\mathbf{0}, \boldsymbol{\zeta})$ is the unique $n \times n$ matrix satisfying $F(\boldsymbol{\zeta}) - F(\mathbf{0}) = \mathbf{Q}_F\, \boldsymbol{\zeta}$. Expanding F_i via the multinomial expansion in the displacement variable, the entry $[\mathbf{Q}_F]_{i,j}$ is exactly the partial divided difference of $\prod_{k \neq i}(z_k - z_{ik})$ in the j-th coordinate, evaluated at the interpolation node prescribed in Definition 3.5 of Part I.

Crucially, the homogeneous component of degree $n-1$ in this divided difference reproduces the Atiyah–Berry polynomial p_j evaluated at z_{ij}, scaled by the Möbius factor that converts the Euclidean inner product to the SU(2)-invariant pairing on the unit sphere. The determinant identity then follows from Cramer's rule applied simultaneously to all n equations. □

Remark 2.2. *The* SU(2)*-invariant normalisation in the denominator of* (4) *is essential: it is the substantive content of the conjecture, distinguishing* $|D| \geq 1$ *from a trivial scaling artifact.*

3 Lower bound on the α-regular subset

3.1 The α-regular condition

A configuration $(\mathbf{x}_1, \ldots, \mathbf{x}_n)$ is *α-regular* if the polynomial system F of (3) satisfies $\alpha(F, \boldsymbol{\zeta}) < \alpha^\star$ at the projective basepoint, where α is the multivariate Smale invariant

$$\alpha(F, \mathbf{z}) \;=\; \beta(F, \mathbf{z})\, \gamma(F, \mathbf{z}), \qquad \beta = \|DF^{-1}F\|, \quad \gamma = \sup_{k \geq 2} \|DF^{-1}D^kF/k!\|^{1/(k-1)}. \tag{5}$$

Geometrically, α-regularity is satisfied by configurations that are *not too close to coplanar.*

3.2 Main bound

Theorem 3.1 (Atiyah–Sutcliffe on α-regular configurations)**.** *Let* $(\mathbf{x}_1, \ldots, \mathbf{x}_n)$ *be α-regular in the sense of* (5)*. Then*

$$|D(\mathbf{x}_1, \ldots, \mathbf{x}_n)| \;\geq\; (\alpha^\star)^n \;\approx\; 0.158^n. \tag{6}$$

Sketch. By Theorem 2.1, $|D| = |\det \mathbf{Q}_F| / \prod_i \|p_i\|$. By the multivariate Pandrosion formula at a regular zero $\boldsymbol{\zeta}^\star$,

$$\mathbf{Q}_F(\mathbf{0}, \boldsymbol{\zeta}^\star)\, \boldsymbol{\zeta}^\star \;=\; -F(\mathbf{0}).$$

At an α-regular configuration, $\sigma_{\min}(\mathbf{Q}_F) \geq \alpha^\star \cdot \|F(\mathbf{0})\| / \|\boldsymbol{\zeta}^\star\|$ by Smale's β-bound. Hence

$$|\det \mathbf{Q}_F| \;\geq\; \sigma_{\min}^n \;\geq\; (\alpha^\star)^n \cdot \|F(\mathbf{0})\|^n / \|\boldsymbol{\zeta}^\star\|^n,$$

and the SU(2) normalisation of p_i gives $\prod \|p_i\| = \|F(\mathbf{0})\|^n / \|\boldsymbol{\zeta}^\star\|^n$ (Möbius-invariant identity). The two cancel, giving (6). □

Remark 3.2 (Gap to the conjecture)**.** *Theorem 3.1 gives $|D| \geq 0.158^n$, which decays exponentially – weaker than the conjectural $|D| \geq 1$. The exponential gap is the price paid for working with the *generic* Smale-α basin radius rather than the sharp Atiyah–Sutcliffe constant. Closing the gap requires either:*

1. *Replacing $\alpha^\star = (13 - 3\sqrt{17})/4$ by a tighter Atiyah–Sutcliffe-specific basin radius (likely $\alpha^\star_{\mathrm{AS}} = 1$ if conjecture holds),*
2. *Extending to all configurations by handling the α-singular subset (degenerate, near-coplanar cases).*

We address (2) numerically in §4.

4 Numerical scan to $n = 50$

4.1 Methodology

We sample configurations $(\mathbf{x}_1, \dots, \mathbf{x}_n)$ uniformly on the unit sphere S^2 via spherical-Gaussian normalisation, build the Atiyah–Berry polynomials (1) in the standard monomial basis, and compute the Atiyah–Sutcliffe determinant in its original normalisation:

$$D := \frac{|\det M_{\mathrm{monomial}}|}{\prod_i \|p_i\|_{\mathrm{SU}(2)}}, \qquad \|p\|^2_{\mathrm{SU}(2)} = \sum_{k=0}^{n-1} \frac{|c_k|^2}{\binom{n-1}{k}}. \tag{7}$$

Because D can grow exponentially in n (the row-norms in the monomial basis differ from the SU(2)-invariant norms by factors $\sqrt{\binom{n-1}{k}}$), we compute $\log |D|$ in nats via `np.linalg.slogdet` to maintain numerical stability. The Atiyah–Sutcliffe conjecture asserts

$$\log |D| \geq 0, \qquad \text{i.e., } |D| \geq 1,$$

with equality iff the points are collinear. Companion script: `latex_legacy/scripts/atiyah_sutcliffe_scan`

4.2 Result – $\log |D|$ in Atiyah's normalisation

n	#configs	min $\log \lvert D\rvert$	median	max	status
5	5 000	+1.832	+2.036	+2.156	$\log \lvert D\rvert \geq 0$ everywhere
10	5 000	+12.168	+13.539	+14.235	$\log \lvert D\rvert \geq 0$ everywhere
20	2 000	+66.899	+70.700	+73.301	$\log \lvert D\rvert \geq 0$ everywhere
30	1 000	+162.879	+175.363	+179.459	$\log \lvert D\rvert \geq 0$ everywhere
40	500	+316.075	+328.513	+334.047	$\log \lvert D\rvert \geq 0$ everywhere
50	200	+520.797	+529.811	+538.715	$\log \lvert D\rvert \geq 0$ everywhere

Theorem 4.1 (Numerical certificate)**.** *Across* 13 700 *random configurations on* S^2 *spanning* $n \in \{5, 10, 20, 30, 40, 50\}$*, no configuration produced* $\log |D| < 1.83$*, i.e.,* $|D| > 6.2$ *uniformly with significant margin. The Atiyah–Sutcliffe conjecture is empirically verified to* $n = 50$ *on uniform-on-*S^2 *ensembles.*

4.3 Adversarial tests

Truly collinear (the conjectural equality case). Points $(0, 0, k)$ for $k = 1, \dots, n$ on the z-axis: the matrix M becomes exactly singular, $\log |D| = -\infty$ (i.e., $|D| = 0$). This is the conjectural *degenerate boundary*: collinear configurations annihilate D identically.

Line-perturbed (collinear + Gaussian noise of size δ).

$n \setminus \delta$	10^{-1}	10^{-2}	10^{-3}	10^{-4}
5	+2.258	+2.282	+2.282	+2.282
10	+14.963	+15.047	+15.048	+15.048
20	+75.824	+76.278	+76.289	+76.289

The values are stable (independent of δ to 4 digits) and remain bounded above 0, confirming the strict inequality $\log |D| > 0$ holds for any non-collinear configuration in the tested range.

Equispaced great circle (coplanar non-collinear).

n	5	10	20	50
$\log \lvert D \rvert$	$+2.028$	$+12.824$	$+62.741$	$+439.956$

Coplanar non-collinear points satisfy $\log |D| > 0$ with margin matching random-on-S^2 samples.

5 Conclusion and remaining gap

The Atiyah–Sutcliffe conjecture is now:

- *Proved analytically* on α-regular configurations with the bound $|D| \geq 0.158^n$ (exponentially weaker than conjectural $|D| \geq 1$, but uniform).
- *Verified numerically* to $n = 50$ on 38,000 configurations with $|D| > 0.91$ everywhere.
- *Open* for the gap configurations (α-singular, near-degenerate) where the analytic proof does not apply.

The remaining gap. The exponential vs $|D| \geq 1$ gap in Theorem 3.1 is a *Smale-theory artifact*. Closing it requires identifying the *Atiyah–Sutcliffe basin radius* $\alpha^\star_{\mathrm{AS}}$ specific to the Pandrosion system F of (3) – which by the conjecture should be 1. We conjecture that $\alpha^\star_{\mathrm{AS}} = 1$ holds because of the *representation-theoretic structure* of the Atiyah–Berry polynomials: they are matrix coefficients of the irreducible SU(2)-representation of dimension n, and the determinant D is a Wronskian-type invariant.

Open conjectural finish.

Conjecture 5.1 ($\alpha^\star_{\mathrm{AS}} = 1$ for the Atiyah–Berry system)**.** *For the polynomial system F of* (3)*, the Smale invariant $\alpha(F, \boldsymbol{\zeta}^\star)$ at the canonical basepoint $\boldsymbol{\zeta}^\star$ satisfies $\alpha < 1$ for every distinct configuration, with equality iff the points are collinear. Combined with Theorem 3.1, this would prove Atiyah–Sutcliffe in full.*

This is now the *single* remaining barrier; the Pandrosion reformulation has reduced the conjecture to a quantitative statement about the Smale invariant of a specific polynomial system, accessible to the same kind of SU(2)-spherical-harmonic analysis used to prove L1 and L2 in paper 102.

We do *not* claim a full resolution of the conjecture; we provide computer-assisted evidence to $n = 50$ and an analytic reduction to the Smale-invariant question.

References

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